Mason Bane

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Summary: Computer Science student focused on the development and integration of high-fidelity C/C++ simulations and analysis tools on Linux. Experience spans the full lifecycle—from optimizing CUDA kernels for real-time, large-scale system simulation (20,000+ nodes) to hardening tooling for validation and deployment. Effective collaborator skilled with Git, CMake, and cross-disciplinary teams. U.S. citizen, clearance-eligible.

Education

Tarleton State University — Stephenville, TX
Bachelor of Science in Computer Science, Concentration in Computer Engineering — *Aug. 2022 – May 2026 (Expected)*
Minor in Mathematics
GPA: 3.94/4.00 (Institutional) — Cumulative: 3.75/4.00
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Architecture, Software Engineering, Parallel Programming, Procedural Programming, C++ Programming, Mathematical Modeling

Hill College — Hillsboro, TX
Associate of Arts in Liberal Arts — *Aug. 2019 – Sept. 2023*

Experience

Undergraduate Research Assistant - Lead Programmer — May 2024 – Present
Tarleton State University — *Stephenville, TX*
- Developed and optimized multiple CUDA/C++ N-body simulations across diverse domains (cardiac arrhythmias, microplastic coagulation), profiling kernels and tuning memory access patterns to achieve real-time performance on lab hardware.
- Led complete migration from legacy OpenGL rendering to modern C++/ImGui framework, delivering an interactive GUI that transformed usability and enabled non-programmers to run complex parameter sweeps independently.
- Collaborated across disciplines to bridge computer science and biomedical research, translating vaious scientific concepts into computational models and simulation requirements.

Undergraduate Technology Specialist - HPC Lab Manager — Aug. 2024 – Present
Tarleton State University — *Stephenville, TX*
- Maintained 15+ Linux systems in production-like research environment, driving 99%+ uptime via proactive patching, hardware diagnostics, and rapid incident response.
- Diagnosed memory, storage, and OS configuration issues; validated fixes and documented runbooks to keep simulation and tooling stacks consistent.
- Standardized configurations with Git-backed docs and scripts, improving onboarding and reducing integration drift across lab nodes.

Projects

N-body Digital Twin of the Left Atrium | *NIH Grant #1R15HL179671-01* | *C, CUDA* — Aug. 2024 – Present
- Engineered parallel CUDA kernels for a 20,000+ node cardiac mesh, modeling electrical propagation and arrhythmia dynamics with near real-time performance for clinical research applications.
- Integrated cardiac physiology models (action potential dynamics, refractory periods, conduction velocities) into computational framework, bridging biomedical science and high-performance computing.
- Built parameter control interface enabling medical researchers to sweep simulation variables (pacing frequency, tissue properties) without C++ knowledge, accelerating model verification and analysis cycles.

N-Body Simulation of Microplastic Coagulation | *C/C++, CUDA, Linux, OpenGL* — May 2024 – Aug. 2024
- Optimized CUDA-based microplastic coagulation simulation; improved kernel performance and validated results against expected aggregation behaviors.
- Developed OpenGL visualization with configurable simulation parameters, enabling rapid experimentation and clear presentation of research findings.
- Designed simulation to mirror lab setups and de-risk wet-lab work by vetting force models, mixing behavior, and parameter ranges in a GPU sandbox before physical experiments.

Technical Skills

Languages: C, C++, Python, Java, Bash, ARM Assembly
Systems & Platforms: Linux, STM32 (Basics)
Modeling, Simulation & Parallel Computing: CUDA, OpenGL, MATLAB, ImGui
Software Engineering Tools: Git, CMake/Make, GDB, VS Code/Visual Studio, Microsoft Office
Practices: Scripting for Build/Test Automation, Validation Workflows, Software Integration